

Project Initialization and Planning Phase — AI BlogNest

Date	17 September 2026
Team ID	SWTID-2026-3342
Project Name	AI BlogNest API
Maximum Marks	5 Marks

The proposal report aims to give content creators and small editorial teams a secure, AI-assisted way to write, organize, and moderate blog content — replacing the blank page and manual moderation with role-aware, generative-AI-backed workflows.

Project Overview

Objective	To provide content teams with a secure, centralized platform to create, organize, and moderate blog content, using Generative AI to draft posts from a topic and summarize existing content, with role-based access for Admins, Editors, Authors, and Readers.
Scope	Deliver a RESTful backend (Node.js / Express / MongoDB) covering user authentication and roles, full blog CRUD with categories/tags/drafts, comment management with moderation, and AI-driven blog generation and summarization via Google Gemini.

Problem Statement

Description	Content creators must write and edit every post manually, summarize long content by hand, and manage comments and contributor permissions with no structured, role-aware system to support them.
Impact	Reduces the time and effort needed to produce and maintain blog content, improves editorial governance through role separation, and demonstrates a scalable pattern for combining a content-management backend with generative AI.

Proposed Solution

Approach	Build a modular backend where Controllers orchestrate blog/comment operations and Gemini AI calls (via a dedicated <code>geminiService.js</code>), while Mongoose Models manage User, Blog, and Comment persistence, secured end-to-end with JWT and role middleware.
Key Features	- JWT authentication with four roles: Admin, Editor, Author, Reader - Full blog CRUD with drafts, publishing status, categories and tags - Comment system with moderation (pending → approved workflow) - AI Blog Generator – produces a structured post from a topic/category - AI Content Summarizer – condenses existing content into a concise summary - Centralized error handling & input validation

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specification (no GPU required – this is a backend REST service, not a model-training workload)	Intel Core i5 / AMD Ryzen 5 or equivalent
Memory	RAM specification	8 GB minimum recommended for running MongoDB, the Node server, and API clients concurrently
Storage	Disk space for code, node_modules, and logs	At least 1 GB available disk space
Software		
Frameworks	Node.js runtime & web framework	Node.js, Express.js
Libraries	Core dependencies	mongoose, cors, morgan, dotenv, Google Gemini AI SDK
Development Environment	IDE / API client	Visual Studio Code, Postman / Thunder Client, nodemon (dev)
Data		
Data	Source, size, format	Google Gemini AI (generative text for blog drafts & summaries); persisted in MongoDB Users, Blogs & Comments collections